

Qualification Pack



Petrochemical Plant Operator Production

QP Code: HYC/Q4501

Version: 1.0

NSQF Level: 4.5

Hydrocarbon Sector Skill Council || OIDB Bhawan, Tower C, 2nd Floor, Plot No. 2, Vikas Marg, Sector - 73

Noida (U.P) -201301 || email: anjalimanhas@hsscindia.in

Qualification Pack

Contents

HYC/Q4501: Petrochemical Plant Operator Production	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
HYC/N4501: Preventive maintenance at petrochemical plant	5
HYC/N4502: Operate, repair and maintenance activities at petrochemical plant	9
HYC/N9301: Working effectively in a team	14
HYC/N9302: Maintain health, safety and security procedures	18
DGT/VSQ/N0102: Employability Skills (60 Hours)	23
Assessment Guidelines and Weightage	30
<i>Assessment Guidelines</i>	30
<i>Assessment Weightage</i>	31
Acronyms	32
Glossary	34

Qualification Pack

HYC/Q4501: Petrochemical Plant Operator Production

Brief Job Description

Individual at this job is responsible for monitoring and controlling the production processes in a petrochemical plant. The person at this job is responsible for safely and efficiently operating the plant to ensure maximum efficiency, cost-effectiveness and quality of the products.

Personal Attributes

The individual should have good sense of responsibility, a good organizer and time keeper. The person should have the ability to work individually / team in different shift.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4501: Preventive maintenance at petrochemical plant](#)
2. [HYC/N4502: Operate, repair and maintenance activities at petrochemical plant](#)
3. [HYC/N9301: Working effectively in a team](#)
4. [HYC/N9302: Maintain health, safety and security procedures](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Midstream, Downstream
Occupation	Petrochemical Plant Operations & Maintenance
Country	India
NSQF Level	4.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO2015/3122.1400

Qualification Pack

Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (in relevant branch) OR 12th grade Pass with 1-2 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-4.5-HY-02188-2024-V1-HSSCI
NQR Version	1.0

Qualification Pack

HYC/N4501: Preventive maintenance at petrochemical plant

Description

This unit is about preventive maintenance to be carried out at petrochemical production plant.

Scope

The scope covers the following :

- This unit/task covers the following:
- Preventive maintenance of instrument and equipment installed at petrochemical plant

Elements and Performance Criteria

Preventive maintenance of instrument and equipment installed at petrochemical plant

To be competent, the user/individual on the job must be able to:

- PC1.** perform routine visual inspections of all plant equipment and systems, including piping, pressure vessels, tanks, pumps, valves, Monitoring of TMT Tube metal temperature etc.
- PC2.** monitor and adjust process parameters according to recommended intervals, process systems for corrosion and erosion
- PC3.** check and adjust operating and safety controls for proper functioning.
- PC4.** inspect and maintain process safety systems, utility piping, pumps, valves, fire protection systems, etc.
- PC5.** maintain plant utilities, such as steam, compressed air, and cooling water systems
- PC6.** report logs, finding and fault to the supervisor/manager

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principle of petrochemical plant operation
- KU2.** standard operating procedures of petrochemical product production
- KU3.** importance of maintaining clean workplace
- KU4.** relevant health and safety requirements applicable to the work place
- KU5.** the importance of personal protective equipment used during welding operation
- KU6.** ways of efficiently managing material and water in the process
- KU7.** the difference between recyclable/reusable and waste material
- KU8.** the concept of Mission Karmayogi
- KU9.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** read and interpret refinery operation blueprint
- GS2.** assess complexity and criticality of task
- GS3.** communicate effectively with team members
- GS4.** perform basic mathematical calculation
- GS5.** plan and organize the marketing activity

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preventive maintenance of instrument and equipment installed at petrochemical plant</i>	20	30	-	-
PC1. perform routine visual inspections of all plant equipment and systems, including piping, pressure vessels, tanks, pumps, valves, Monitoring of TMT Tube metal temperature etc.	4	6	-	-
PC2. monitor and adjust process parameters according to recommended intervals, process systems for corrosion and erosion	4	6	-	-
PC3. check and adjust operating and safety controls for proper functioning.	4	6	-	-
PC4. inspect and maintain process safety systems, utility piping, pumps, valves, fire protection systems, etc.	4	6	-	-
PC5. maintain plant utilities, such as steam, compressed air, and cooling water systems	2	3	-	-
PC6. report logs, finding and fault to the supervisor/manager	2	3	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4501
NOS Name	Preventive maintenance at petrochemical plant
Sector	Hydrocarbon
Sub-Sector	
Occupation	Petrochemical Plant Operations & Maintenance
NSQF Level	4.5
Credits	6
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

HYC/N4502: Operate, repair and maintenance activities at petrochemical plant

Description

This unit is about operating, repair and maintenance activities for proper functioning of petrochemical production plant.

Scope

The scope covers the following :

- This unit/task covers the following:
- Operate different component at petrochemical production plant
- Analyse equipment and instruments for repairing or maintenance at a petrochemical production plant

Elements and Performance Criteria

Operate different component at petrochemical production plant

To be competent, the user/individual on the job must be able to:

- PC1.** operate pumps, valves, reactors, heat exchangers, turbines, compressors, separators, storage tanks, piping, etc. and follow all safety protocols and procedures
- PC2.** regulate pumps and compressors as per the instruction of supervisor/manager
- PC3.** monitor and adjust production process parameters such as pressure, temperature, flow, and levels
- PC4.** control the plant operations such as controlling the various processes and operations within the plant and controlling the temperature, pressure and other parameters for the various processes.
- PC5.** read temperature in pressure gauges, flowmeters, vacuum and time, record readings in the log
- PC6.** monitor and operate processing unit, cracking unit, separation unit, synthesis unit, purification unit, product packaging unit, etc. to produce petrochemical product.
- PC7.** clean processing units' interior by circulating solvent and chemicals as per SOP
- PC8.** follow all shift-specific operational procedures and protocols
- PC9.** report logs, finding and fault to the supervisor/manager
- PC10.** Assisting in Shutdown activity like preparing blinding list, adhering to safe shutdown procedure, SOP etc.
- PC11.** Assisting in Startup activity like leak checking, gasket checking, loop checking etc.

Analyse equipment and instruments for repairing or maintenance at a petrochemical production plant

To be competent, the user/individual on the job must be able to:

- PC12.** identify the repair and maintenance need in the plant
- PC13.** schedule and inform repair and maintenance activity at plant

Qualification Pack

- PC14.** apply solution for maintenance/repair on faulty parts such as control system/electrical system/ mechanical system, etc.
- PC15.** check all the inspection points as per maintenance/repair checklist
- PC16.** test, visually inspect or using calibration instruments and tools to ensure proper functioning of repaired instrument/component
- PC17.** utilize the material required during repair and maintenance activity efficiently to minimize the waste and store the unused or reusable material
- PC18.** clean the site and remove the maintenance /repair signages from the site
- PC19.** inform all concerns after repair / maintenance about site readiness
- PC20.** maintain/prepare maintenance/repair log as per the SOP
- PC21.** prepare reports and documentation related to repair / maintenance in the Plant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principle of refinery operation
- KU2.** standard operating procedures for refinery operation
- KU3.** importance of maintaining clean workplace
- KU4.** relevant health and safety requirements applicable to the work place
- KU5.** the importance of personal protective equipment used during welding operation
- KU6.** ways of efficiently managing material and water in the process
- KU7.** the difference between recyclable/reusable and waste material
- KU8.** the concept of Mission Karmayogi
- KU9.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret refinery operation blueprint
- GS2.** assess complexity and criticality of task
- GS3.** communicate effectively with team members
- GS4.** perform basic mathematical calculation
- GS5.** plan and organize the marketing activity

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate different component at petrochemical production plant</i>	23	28	-	-
PC1. operate pumps, valves, reactors, heat exchangers, turbines, compressors, separators, storage tanks, piping, etc. and follow all safety protocols and procedures	2	3	-	-
PC2. regulate pumps and compressors as per the instruction of supervisor/manager	2	3	-	-
PC3. monitor and adjust production process parameters such as pressure, temperature, flow, and levels	2	3	-	-
PC4. control the plant operations such as controlling the various processes and operations within the plant and controlling the temperature, pressure and other parameters for the various processes.	2	3	-	-
PC5. read temperature in pressure gauges, flowmeters, vacuum and time, record readings in the log	2	3	-	-
PC6. monitor and operate processing unit, cracking unit, separation unit, synthesis unit, purification unit, product packaging unit, etc. to produce petrochemical product.	2	2	-	-
PC7. clean processing units' interior by circulating solvent and chemicals as per SOP	3	3	-	-
PC8. follow all shift-specific operational procedures and protocols	2	2	-	-
PC9. report logs, finding and fault to the supervisor/manager	3	3	-	-
PC10. Assisting in Shutdown activity like preparing blinding list, adhering to safe shutdown procedure, SOP etc.	2	2	-	-
PC11. Assisting in Startup activity like leak checking, gasket checking, loop checking etc.	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse equipment and instruments for repairing or maintenance at a petrochemical production plant</i>	17	32	-	-
PC12. identify the repair and maintenance need in the plant	1	3	-	-
PC13. schedule and inform repair and maintenance activity at plant	2	4	-	-
PC14. apply solution for maintenance/repair on faulty parts such as control system/electrical system/ mechanical system, etc.	1	3	-	-
PC15. check all the inspection points as per maintenance/repair checklist	2	4	-	-
PC16. test, visually inspect or using calibration instruments and tools to ensure proper functioning of repaired instrument/component	2	4	-	-
PC17. utilize the material required during repair and maintenance activity efficiently to minimize the waste and store the unused or reusable material	1	3	-	-
PC18. clean the site and remove the maintenance /repair signages from the site	2	3	-	-
PC19. inform all concerns after repair / maintenance about site readiness	2	3	-	-
PC20. maintain/prepare maintenance/repair log as per the SOP	2	3	-	-
PC21. prepare reports and documentation related to repair / maintenance in the Plant	2	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4502
NOS Name	Operate, repair and maintenance activities at petrochemical plant
Sector	Hydrocarbon
Sub-Sector	
Occupation	Petrochemical Plant Operations & Maintenance
NSQF Level	4.5
Credits	7
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4.5
Credits	1.5
Version	6.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment

Scope

The scope covers the following :

- This unit/task covers the following:
- Practice health and safety measures
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** identify documents, location and people responsible for health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** identify common safety signs, displayed in various areas

Follow fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC7.** use the various appropriate fire extinguishers on different types of fires correctly
- PC8.** follow rescue techniques applied during fire hazard
- PC9.** follow good housekeeping practice in order to prevent fire hazards
- PC10.** list issues concerning the safety in work place
- PC11.** inform fire safety department about any near-miss incidents in the work place
- PC12.** follow the applicable laws, regulations and codes as per safety standard
- PC13.** prepare written accident/incident report and share with the concerned officer/department

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC14.** provide appropriate first aid to victims in emergency situation
- PC15.** demonstrate basic techniques of bandaging
- PC16.** respond promptly and appropriately to an accident

Qualification Pack

- PC17.** Use various rescue equipment & techniques and perform rescue activity during an accident in real or simulated environments
- PC18.** demonstrate correct method to rescue injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks during scaffolding work
- KU5.** health condition of individual as well as health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** working at heights regulations and best practices
- KU9.** various causes of fire
- KU10.** techniques of using different types of fire extinguishers
- KU11.** different materials used for extinguishing fire
- KU12.** various types of safety signs and their significance
- KU13.** various types of rescue equipment and techniques
- KU14.** precautions to be taken to avoid suspension trauma while rescuing a person hanging from harness

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	1	4	-	-
PC2. identify documents, location and people responsible for health and safety in the workplace	2	1	-	-
PC3. identify possible causes of risk or accident in the workplace	1	3	-	-
PC4. carry out safe working practices while dealing with hazards to ensure the safety of self and others	2	2	-	-
PC5. lift heavy objects safely using correct procedures	1	3	-	-
PC6. identify common safety signs, displayed in various areas	2	2	-	-
<i>Follow fire safety procedures</i>	12	16	-	-
PC7. use the various appropriate fire extinguishers on different types of fires correctly	2	2	-	-
PC8. follow rescue techniques applied during fire hazard	2	2	-	-
PC9. follow good housekeeping practice in order to prevent fire hazards	2	2	-	-
PC10. list issues concerning the safety in work place	1	3	-	-
PC11. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC12. follow the applicable laws, regulations and codes as per safety standard	1	3	-	-
PC13. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow emergencies, rescue and first-aid procedures</i>	3	15	-	-
PC14. provide appropriate first aid to victims in emergency situation	1	4	-	-
PC15. demonstrate basic techniques of bandaging	1	4	-	-
PC16. respond promptly and appropriately to an accident	-	3	-	-
PC17. Use various rescue equipment & techniques and perform rescue activity during an accident in real or simulated environments	1	4	-	-
PC18. demonstrate correct method to rescue injured people and others during an emergency	-	-	-	-
NOS Total	24	46	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	3.5
Credits	1.5
Version	5.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4501.Preventive maintenance at petrochemical plant	20	30	-	-	50	20
HYC/N4502.Operate, repair and maintenance activities at petrochemical plant	40	60	-	-	100	30
HYC/N9301.Working effectively in a team	20	30	-	-	50	20
HYC/N9302.Maintain health, safety and security procedures	24	46	-	-	70	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	124	196	-	-	320	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills

Qualification Pack

SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.